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Task Description:	The space radiation environment poses a range of potential health risks to astronauts. Exposure to high-energy protons and heavy ions within galactic cosmic rays or solar particle events can damage critical cellular targets leading to a number of deleterious effects. Space travel has also been found to elicit changes in the redox state of cells that may exacerbate these adverse effects. These risks may be particularly problematic to stem/precursor cells within the CNS, where exposure to irradiation may promote genomic instability which could result in cancer and/or cognitive dysfunction. In vitro and in vivo we have shown neural precursor cells are exquisitely sensitive to irradiation, undergoing a dose-dependent induction of apoptosis that is associated with a chronic oxidative stress. These effects are strongly correlated with a persistent inhibition of neurogenesis that may underlie cognitive deficits observed after cranial irradiation. We have also shown that human brain tumors contain a significant number of cells that express many of the same immature lineage-specific markers common to neural precursor cells, supporting a possible role for these cells in the development of brain tumors. Based on these observations, we hypothesize that neural precursor cells exposed to high energy ions will be particularly susceptible to genomic instability and in a manner that depends upon redox state. Genomic instability is a process believed to drive cancer progression, and recently, redox state has been implicated in the induction and perpetuation of genomic instability. Redox state has been found to be a critical determinant in regulating the response of the CNS to a variety of damaging treatments, including irradiation. Changes in the redox environment are often associated with alterations in cellular metabolism which can promote oxidative stress, and these types of changes are common to tumors including those within the CNS. Because exposure to radiation has been found to cause genomic instability, and since oxidative stress is associated with the radiation response of neural precursor cells, it follows that changes in redox state might constitute a biochemical mechanism for modifying the radioresponse of the CNS. We are proposing a series of studies to test the overall hypothesis that alterations in cellular redox state promote genomic instability that can adversely impact the CNS upon exposure to high LET radiation. While the development of genomic instability may predispose neural precursor cells to transformation events in the CNS, the unstable phenotype may also lead to radioresistance that changes the phenotypic fate of neural precursor cells that impairs neurogenesis. We propose that changes in redox state play a contributory if not causative role in regulating the radiation response of neural stem cells in vitro and in vivo; changes that will foster the onset of radiation-induced genomic instability, radioresistance and altered neurogenesis. Using an in vitro and in vivo approach the goals of our project will be addressed in the following 3 specific aims: Specific aim 1: Elucidate the dose-response relationship for the induction of genomic instability in neural precursor cells cultured in vitro or isolated from mouse brain after exposure to protons and iron ions. Specific aim 2: Determine if chronic changes in cellular redox state alter the dose response relationship determined above and predispose neural precursor cells to proton- and iron ion-induced genomic instability. Specific aim 3: Determine if differentiation of genomically unstable neural precursor cells is altered in vitro, or if neurogenesis is altered in vivo when unstable precursor cells are implanted into the brains of control or irradiated mice. Task Objective: To determine whether changes in redox state adversely impact the CNS. We will assess whether oxidative stress alters the radiosensitivity of neural precursors and predisposes them radiation-induced genomic instability. Experiments will ascertain whether such changes alter precursor cell plasticity in a manner that compromises their normal function. These studies will promote an understanding of the space radiation effects on the CNS. Data collected from these studies will improve both our understanding of the basic biology underlying CNS radiation effects, and in assessing risks to astronauts engaged in space travel. This work will establish the extent to which redox related changes derived from environmental stressors (oxidative injury) and/or damaging agents (ionizing radiation) are problematic to astronauts and whether specific countermeasures can be developed to minimize such risk.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will shed light on many of the basic biological processes that are associated with exposure of the CNS to ionizing radiation. As such, our work is directly relevant to those involved in the treatment of CNS tumors via radiotherapy. Such individuals include clinicians, basic scientists and even patients, all of whom have a vested interest in minimizing side-effects associated with the exposure of normal tissue to ionizing radiation. Outside of the clinical relevance, our work should foster a general interest to those members of the public supporting the long-term goals of NASA involving the expoloration and colonization of space.
	Summary of Research Scientific Objectives: We are proposing a series of studies to test the overall hypothesis that alterations in cellular redox state promote genomic instability that can adversely impact the CNS upon exposure to high LET radiation. While the development of genomic instability may predispose neural precursor cells to transformation events in the CNS, the unstable phenotype may also lead to radioresistance that changes the phenotypic fate of neural precursor cells that impairs neurogenesis. We propose that changes in redox state play a contributory if not causative role in regulating the radiation response of neural stem cells in vitro and in vivo; changes that will foster the onset of radiation-induced genomic instability, radioresistance and altered neurogenesis. Using an in vitro and in vivo approach the goals of our project will be addressed in the following 3 specific aims: Specific aim 1: Elucidate the dose-response relationship for the induction of genomic instability in neural precursor cells cultured in vitro or isolated from mouse brain after exposure to protons and iron ions. Specific aim 2: Determine if chronic changes in cellular redox state alter the dose response relationship determined above and predispose neural precursor cells to proton- and iron ion-induced genomic instability. Specific aim 3: Determine if differentiation of genomically unstable neural precursor cells is altered in vitro, or if neurogenesis is altered in vivo when unstable precursor cells are implanted into the brains of control or irradiated mice. Summary of Accomplishments: We have made significant progress during the initial phases of our funding period. Our recent findings now support a major hypothesis of our overall grant, namely, that oxidative stress changes radiosensitivity. We proposed a number of strategies directed at altering the redox state of cells. We have now been able to successfully manipulate the level of oxidative stress in neural precursor cells using three distinct methodologies, and have determined that increased oxidative stress leads to an enhancement of radiosensitivity. Specifically, we have elevated oxidative stress through endogenous manipulation of metabolism, through exogenous agents, or through perturbation of mitochondrial electron transport, and have in each instance, been able to increase the sensitivity of cells

Task Progress:

to ionizing radiation. We have also completed initial work at Loma Linda University where we have been able to demonstrate that oxidative stress also enhances the sensitivity of neural precursors to 250 MeV proton irradiation. Given our positive results from these initial studies, we can now proceed to determine how oxidative stress impacts genomic instability, differentiation and neurogenesis in the CNS.

Major accomplishments are summarized below: 1) In vitro studies using hippocampal precursors have revealed that endogenous oxidative stress increases the sensitivity of these cells to X-rays. 2) In vitro studies using hippocampal precursors have revealed that exogenous oxidative stress increases the sensitivity of these cells to X-rays and to 250 MeV protons. 3) In vitro studies have shown that perturbation of mitochondrial electron transport increases oxidative stress and enhances the sensitivity of neural precursors to irradiation. 4) We have 3 published manuscripts, and one in preparation. Details of our findings are presented below. Endogenous manipulation of oxidative stress: One of our first, and most critical objectives in this grant was to determine whether alterations in redox state changed the radiosensitivity of neural precursors. To investigate this we took advantage of our recent work showing dramatic changes in oxidative stress that depended on neural precursor cell density (see Limoli et. al. PNAS 2004). This work provided a simple yet elegant avenue for manipulating endogenous levels of oxidative stress. Consequently, we wanted to re-confirm the differences in the levels of reactive oxygen species (ROS) previously described between low-density (LD: <104/cells/cm2) and high-density (HD: >105 cells/cm2) cultures (Limoli et. al. PNAS 2004). Cells seeded at LD and HD and assessed for differences in ROS showed that LD cultures exhibited increased levels of intracellular ROS when compared to their HD counterparts. These results corroborated our past findings (Limoli et. al. PNAS 2004). To more thoroughly examine the potential consequences imparted by these endogenous changes in ROS we sought to determine whether increased oxidative stress was associated with an increased radiosensitivity and/or inhibition of growth. Cells cultured at LD or HD were irradiated and reseeded at equal densities. At various times afterwards cells were counted to assess survival. In the absence of irradiation, the consequences of high ROS are evident as cells grown under LD exhibit significant growth delay compared to those cells grown at HD. However, one can also see that when cells grown under either condition are subjected to irradiation, additional growth delay occurs. Additional analysis shows that oxidative stress enhances growth delay by about 1day. Thus, cells subjected to endogenous oxidative stress undergo an inhibition of growth that is enhanced after irradiation. Since growth at different cell densities prior to irradiation clearly impacted oxidative stress and subsequent growth, we sought to determine whether similar effects could be obtained when the number of cells seeded after irradiation was lowered 5-10 fold. Results from these studies corroborated data shown above, and indicated that when the total number of cells was lowered an enhanced inhibition of growth was still evident when cells were irradiated under conditions of oxidative stress (i.e LD conditions). To provide an alternate measure of survival, cells were also assessed for viability using the XTT assay. Since only viable cells can convert (via mitochondrial dehydrogenases) the soluble XTT salt into a colored substrate, absorbance readings can be used to quantify survival. When cells were treated as described above and analyzed by XTT assay very similar results are obtained, confirming that the XTT assay provides a reasonable measure of the number of viable cells present 4-days after irradiation. Similar experiments were also run on cells grown at LD or HD, but irradiated at a lower dose (i.e. 2 Gy). Irradiation under these conditions of endogenous oxidative stress still enhanced the inhibition of growth observed 4 days later. Collectively data suggests that cells subjected to endogenous oxidative stress exhibit an enhanced inhibition of growth after irradiation. Exogenous manipulation of oxidative stress: Since our data showed that endogenous oxidative stress could impair growth we sought to determine whether similar changes might be found by increasing oxidative stress using an exogenous agent. We also sought to determine whether neural precursors might show a different response to protons versus ?-rays, and have since completed a comprehensive series of studies at Loma Linda, where we have optimized conditions for treating neural precursors to chronic oxidative stress. For this work, we relied on our previous work with H2O2, where we have shown that successive or “chronic” treatment regimes to be more efficacious a eliciting a number of altered phenotypes compared to single acute treatments. Thus we investigated whether a similar approach in which either 10 or 20µM H2O2 was added directly to cells once daily over the course of 3 or 5 days could sensitize cells to the effects of ionizing radiation. Cells subjected to these conditions were then exposed to 1-5 Gy of 250 MeV protons and analyzed 3 days later using the XTT assay. Oxidative treatments of unirradiated cells showed these levels of H2O2 to be relatively non-toxic, reducing XTT absorbance by only 16% (0.46/0.55) when compared to untreated controls. While the XTT assay was not particularly sensitive to higher proton doses, there was however a modest dose-response observed for cells not subjected to H2O2 treatment. However, for those cells subjected H2O2, XTT absorbance was consistently reduced to a greater extent when compared to cells not subjected the exogenous oxidant. Since the difference between cells treated with 10 or 20 µM H2O2 were not statistically different, these data have been averaged and plotted as a combined treatment. While extending oxidative treatments an additional 2-days (5 days total) gave qualitatively similar results, longer duration treatments did not afford a further enhancement in sensitivity to irradiation. By normalizing the change observed between untreated and H2O2 treated cells at each proton dose, the relative sensitization afforded by each treatment can be compared. While the relative change shows some variability between doses, in each instance there is a statistically significant difference indicating increased sensitivity to irradiation in all treatment groups. Similar experiments were also run in which cells subjected to chronic oxidative stress were exposed to ?-rays. When this data is analyzed as described above, similar trends are observed indicating that oxidative stress enhances the sensitivity of cells exposed to ionizing radiation. Collectively the foregoing results suggest that endogenous and exogenous increases in oxidative stress enhance the sensitivity of neural precursors to ionizing radiation.

Mitochondrial manipulation of oxidative stress: The results above suggest that oxidative stress leads to enhanced radiosensitivity. While the manipulation of endogenous or exogenous oxidative stress are well worked out, culturing cells under different densities or treating them continuously with exogenous agents for 2 or more days presents certain logistical difficulties for executing these experiments at BNL. Thus in efforts to both develop a more efficient strategy for manipulating oxidative stress and to address potential mechanisms we have optimized another method for tightly controlling oxidative stress through the use of mitochondrial (mt) electron transport chain (ETC) poisons. We have selected two inhibitors that target complex I (menadione) or III (antimycinA) of the mt ETC. These two sites are expected to generate the highest levels of superoxide (ROS) leakage upon inhibition since they possess the highest redox potentials of the mt ETC complexes. Initial studies sought to validate the activity of these agents, and analyzed ROS levels in cells treated with each of these agents using two ROS sensitive dyes. Cells treated with menadione and/or antimycinA were analyzed by FACS over a 2h interval after loading with either CM-DCFDA or mitosox dyes. Each of these dyes becomes fluorescent upon intracellular oxidation and yield qualitatively similar data proportional to the level of ROS. When normalized to untreated controls (set to unity) the data clearly show that these agents elicit rapid and significant elevations in ROS. ROS levels are observed to rise and stay elevated over 2h. The drop in CM-DCFDA fluorescence is likely due to efflux of the dye, which is less problematic for the mitosox analog since it associates more closely with the mitochondrial organelle. Similar data can be derived for the use of menadione, which leads to marked

	and persistent increases in ROS over a 2h post-treatment interval. Lastly, additional studies were conducted at different concentrations and confirmed that each agent was able to elicit ROS over longer times (= 6h) post-treatment. The foregoing studies indicate the utility of using mt ETC inhibitors to control oxidative stress in cells. We have now undertaken preliminary experiments using this approach to determine whether cells subjected to increased mt-derived ROS show altered radiosensitivity. The results of these studies show that similar to endogenous and exogenous treatments detailed above, the increase in ROS derived from mt ETC inhibitors also enhance the sensitivity of cells to irradiation. This new data is again consistent with prior results and corroborates the idea that oxidative stress modifies the radioresponse of neural precursors. In summary we have compiled a wealth of evidence using three distinct strategies that show the same fundamental trend, namely, that increased oxidative stress in neural precursor cells is associated with an enhanced sensitivity to ionizing radiation. These results fit nicely with other observation from our lab that implicates the importance of redox state in regulating the radiation response of neural precursor cells. This line of investigation suggests that redox related issues in the CNS must be fully addressed to understand and minimize the CNS risks associated with exposure to the space radiation environment.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
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